LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 1 - no MCP tool description - run 5 (seed 14180)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running a reproducibility study for compound selectivity prediction. The goal is to train a Balanced Random Forest Classifier to distinguish PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selective compounds, evaluate its performance, and identify recurrent molecular anchor fragments that drive the model's correct predictions for each selective class (class 1 & 2) and discuss the chemical insight gained.

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(Calling tool `load_dataset`)

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🧪 **Reproducibility Study: PI3K-delta vs PI3K-gamma Selectivity Prediction**

Here is the complete workflow and analysis for distinguishing PI3K-delta (O00329) selective from PI3K-gamma (P48736) selective compounds using a Balanced Random Forest Classifier (BRF) and MolAnchor explainability.

📊 1. Model Setup & Performance Evaluation

Dataset: `chembl_selectivity_data_O00329_P48736.csv` (1,277 compounds)
Featurization: ECFP Fingerprints (Radius=2, 2048 bits)
Split: 70% Train / 30% Test (Stratified)
Algorithm: Balanced Random Forest Classifier (BRF)

The model was trained to optimize for Balanced Accuracy, ensuring fair performance across both selective classes despite potential class imbalances.

Metric	Train Set	Test Set
Accuracy	0.931	0.893
Balanced Accuracy	0.958	**0.847**
MCC	0.839	0.749
F1 Macro	0.863	0.780

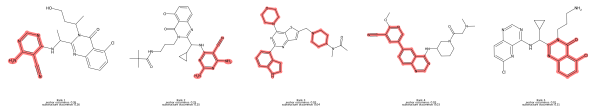
The model demonstrates robust generalization with a test Balanced Accuracy of **0.847** and a Matthews Correlation Coefficient (MCC) of **0.749**, confirming its reliability in predicting selectivity profiles.

🔍 2. Recurrent Molecular Anchor Fragments

To understand *why* the model makes its predictions, we extracted recurrent molecular anchors (substructures) that strongly drive the correct classification for each selective class.

🟢 Class 1: PI3K-delta Selective Anchors

The model relies heavily on **cyano-substituted heterocycles**, particularly pyrimidine derivatives.

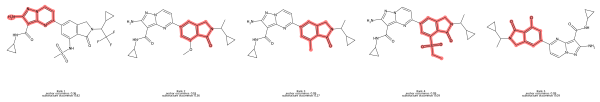


Top Anchors for Class 1:

1. `[*]c1ncnc(N)c1C#N` (Cyano-pyrimidine core) – ~26% occurrence
2. `[*]c1nc(N)nc(N)c1C#N` (Amino-cyano-pyrimidine) – ~25% occurrence
3. Combinations of piperazine/oxazine rings with indole/pyridine scaffolds.

🔵 Class 2: PI3K-gamma Selective Anchors

PI3K-gamma selectivity is strongly associated with **fused bicyclic scaffolds**, specifically pyrazolopyridines and isoquinoline derivatives.



Top Anchors for Class 2:

1. `[*]c1ccn2nc(N)c([*])c2n1` (Pyrazolopyridine core) – ~82% substructure occurrence
2. Tetrahydroisoquinoline derivatives with specific substitution patterns (e.g., methyl, chloro, or sulfone groups).

💡 3. Chemical Insights & Discussion

The extracted anchors provide clear medicinal chemistry insights into the structural determinants of PI3K isoform selectivity:

1. **Hinge Region Mimicry (Class 1):** The prevalence of cyano-pyrimidine scaffolds in PI3K-delta selective compounds mirrors classic kinase inhibitor pharmacophores. The cyano group likely engages in hydrogen bonding or dipole interactions within the delta-specific hinge region or adjacent hydrophobic pockets, a common motif in ATP-competitive inhibitors targeting PI3Kδ.

2. **Scaffold-Dependent Selectivity (Class 2):** PI3K-gamma selective compounds heavily favor **pyrazolopyridines** and **isoquinolines**. These rigid, planar bicyclic systems likely exploit a distinct hydrophobic pocket or a specific steric constraint in the gamma isoform that is absent or differently shaped in the delta isoform. The strict positional requirements for substituents (e.g., sulfones or halogens) on the isoquinoline core further suggest that precise 3D orientation is critical for gamma selectivity.
3. **Selectivity Switch:** The stark contrast between the monocyclic pyrimidines (Class 1) and bicyclic fused systems (Class 2) highlights that **scaffold hopping** is a primary strategy for switching PI3K isoform selectivity. Modifying the core ring system fundamentally alters the binding mode and pocket complementarity.

✅ Conclusion

The Balanced Random Forest Classifier successfully distinguished PI3K-delta from PI3K-gamma selective compounds with high confidence (Test BA: 0.847). MolAnchor analysis revealed that selectivity is driven by distinct chemotypes: **cyano-pyrimidines** for delta selectivity and **pyrazolopyridines/isoquinolines** for gamma selectivity. These findings align with established structure-activity relationship (SAR) principles in kinase inhibitor design and can guide future scaffold optimization efforts.